Code-Layout, Readability and Reusability

Date	01 October 2026
Team ID	SWTID-2026-1940
Project Name	Team Enrollment Confirmation - EduGenie: Google Gemini Powered Learning Assistant
Team Members	R.Priya dharshini, S.Shobana, S.Rakshana, E.Shylaja, S.Sheeba
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the planned codebase for structure, readability, maintainability and reuse. The checklist is aligned with the EduGenie learning-assistant project and can be used during code review to confirm that naming, modular design, comments, error handling and file organization remain consistent.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs throughout the code	Yes	Use one indentation standard across all files.
2	Proper File Structure	Files and folders are logically organized	Yes	Keep UI, backend, AI integration and testing responsibilities separated.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Use descriptive names for enrollment, learner and Gemini-related data.
4	Function / Method Names	Functions are descriptively named	Yes	Use action-oriented names such as create_enrollment(), generate_response().
5	Code Comments	Inline and block comments explain logic	Partial	Document non-obvious Gemini prompts, API handling and validation rules.
6	Modular Design	Code is split into reusable functions/modules	Yes	Keep enrollment, AI, learning and utility logic modular.
7	No Redundant Code	Duplicate or unused code is removed	Partial	Review repeated validation, formatting and API-call logic before final submission.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Validate user input and handle API/network failures with clear messages.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Team Enrollment & Confirmation Module	Frontend + Backend	Enrollment form, validation and confirmation flow	High
2	Gemini AI Integration Module	Gemini API / Backend	Q&A;, explanations and learning assistance	High
3	Learning Content Module	Backend / AI	Concept explanation, summaries and study support	High
4	Quiz / Assessment Module	Backend + UI	Practice questions, evaluation and feedback	High
5	Validation, Error Handling & Utility Module	Backend / Utility	Shared input validation, API errors and common helpers	High

Team Work Assignment

Work is assigned in the same order as the names provided. Each member owns a primary area while supporting integration and final review as required.

S.No	Team Member	Primary Responsibility	Key Deliverables
1	R.Priya dharshini	Frontend & Enrollment Confirmation	Enrollment UI, confirmation screen, form validation presentation, responsive layout and UI integration.
2	S.Shobana	Backend & API Integration	Backend routes/services, enrollment data handling, API integration and secure request/response processing.
3	S.Rakshana	Gemini AI & Q&A; Module	Gemini prompt integration, question-answer flow, contextual responses and AI error handling.
4	E.Shylaja	Learning Modules & Content Features	Learning explanations, summaries/learning support, quiz-related content flow and reusable learning components.
5	S.Sheeba	Testing, Integration & Documentation	Functional testing, integration checks, test cases, bug tracking, README/project documentation and final quality review.

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Clear separation of frontend, backend, AI and utility responsibilities is recommended.
Readability	4	Use descriptive names, consistent indentation and concise comments for complex logic.
Reusability	4	Shared validation, AI service calls and learning components should be implemented as reusable modules.
Documentation / Comments	4	Maintain setup instructions, API configuration notes, module descriptions and important logic comments.
Overall Score	4 / 5	Good planned structure; final rating should be confirmed after complete code review and testing.

Submission Note

The ratings and checklist entries above are a prepared project-review draft. Before submission, the team should verify each item against the actual repository/code and update any Yes/Partial status that differs from the implemented code.

Team Lead / Representative	Faculty / Reviewer
Name: _____	Name: _____
Signature: _____	Signature: _____